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# Boosting Lignin Dissolution and Biomass Fractionation by Innovative Alkaline Deep Eutectic Solvents

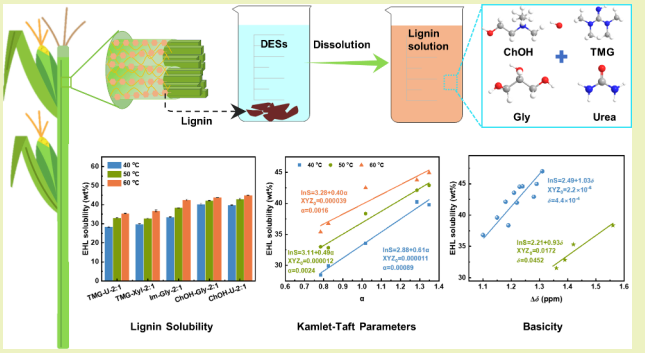
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**ABSTRACT:** Utilization of deep eutectic solvents (DESs) to extract lignin is a highly attractive strategy for biomass upgrading. Although acidic DESs have been extensively investigated, alkaline DESs (ADESs) have rarely been applied in lignin extraction. Herein, several promising ADESs were prepared for lignin dissolution and biomass pretreatment. Lignin solubility showed a good positive correlation with the Kamlet−Taft parameters and the basicities of the prepared ADESs. Notably, the highest solubility of alkali lignin (AL), enzymatically hydrolyzed lignin (EHL), kraft lignin (KL), and dealkaline lignin (DAL) could reach 49.85, 49.46, 53.14, and 53.17 wt % in ADESs at 60 °C, respectively. Importantly, the prepared ADESs could efficiently remove lignin (80.9−97.8%) from corn stover at 120 °C, significantly improving the enzymatic efficiencies of the residues (71.5−85.4%). Besides, lignin with relatively intact structures or lignin fragments with low molecular weight and good dispersion could be successfully generated. These results confirmed the great potential of ADESs for biomass upgrading.




**KEYWORDS:** *alkaline deep eutectic solvents, lignin dissolution, biomass pretreatment, Kamlet−Taft parameters, basicity*

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## INTRODUCTION

Lignocellulosic biomass is mainly composed of lignin, cellulose, and hemicellulose.[1,2] As the most abundant renewable carbon resource globally, lignocellulosic biomass has been considered an ideal substitute for fossil carbon resources. However, the intricate physicochemical structure of lignocellulosic biomass is an obstacle to its high-value utilization.[3] Besides, the products produced from the transformation of the three components are extremely different. For example, cellulose and hemicellulose are commonly employed to produce bioethanol and furfural,[4−6] while lignin, the most prevalent renewable aromatic polymer, is generally used to synthesize aromatic chemicals and fuel products.[7,8] For effective utilization of lignocellulosic biomass, it is essential to fractionate lignocellulosic biomass via a pretreatment process using suitable solvents.[9−11]

Currently, various pretreatment methods have been developed for the fractionation of lignocellulose, including dilute acid treatment, hydrothermal treatment, alkali treatment, *etc.*[12] Industrially, the most commonly used method is kraft pulping. However, non environmentally friendly reagents (*e.g.*, NaOH and $Na_2S$) are generally indispensable for the pulping process, and large amounts of black liquor containing lignin and inorganic salts are generated.[13] In this context, green solvents have gained significant interest in the field of lignocellulose fractionation. Ionic liquids (ILs) are promising solvents for biomass pretreatment owing to their favorable characteristics.[14] Nevertheless, certain drawbacks (*e.g.*, complicated preparation process, poor biodegradability, and high cost) significantly limited the large-scale applications of ILs.[15] In comparison, deep eutectic solvents (DESs), formed through hydrogen-bonding interactions between hydrogen-bond acceptors (HBAs) and hydrogen-bond donors (HBDs), possess obvious advantages, including low cost, easy preparation, mostly nontoxicity, adjustable properties, and high selectivity for lignin extraction.[16−24] These features afford DESs as an emerging class of green solvents for lignocellulose fractionation.

Previous investigations have revealed that acidic DESs, especially those based on choline chloride (ChCl) and lactic acid (LA), possessed strong capabilities for the fractionation of lignocellulose and the dissolution of various type of lignin.[25] For instance, LA-based DESs exhibited strong abilities to dissolve lignin, and the solubility of enzymatically hydrolyzed lignin (EHL) could reach 48.5 wt % at 70 °C.[26] The DESs composed by ChCl and LA could efficiently fractionate bamboo for the production of lignin-containing cellulose nanofibers (LCNF),

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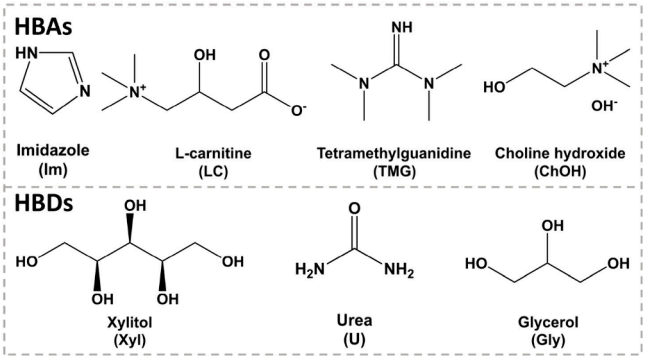
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lignin nanoparticles (LNPs), and carbon quantum dots (CDs).[27] Meanwhile, this type of DESs could be applied for the pretreatment of eucalyptus wood, significantly improving the glucose yield to 94.3% in subsequent cellulase hydrolysis processes.[28] Besides LA-based DESs, acidic DESs prepared using other carboxylic acids as HBDs, *e.g.*, formic acid,[29] propionic acid,[30] oxalic acid,[31] and levulinic acid,[32] could also be employed for efficient lignin removal. In spite of these achievements, most of the DESs developed for lignocellulosic fractionation are still acidic or neutral.

Considering that traditional kraft pulping is an alkali process, alkaline DESs (ADESs) are probably efficient solvents for lignocellulose fractionation. To date, several attempts to use ADESs for lignocellulose fractionation have been conducted. For example, the ADESs composed by ethanolamine (EA) and ChCl could successfully pretreat xylose residues to obtain high delignification rate (93.98%) and enhance enzymatic saccharification (a glucose yield of 90.12%).[33] Moreover, cellulose could be efficiently separated from lignocellulosic biomass using the ADESs formed from glycerol (Gly) and potassium carbonate ($K_2CO_3$), and the maximum cellulose content of the resulting pulp could reach 73.8%.[34] However, the types of efficient ADESs for lignocellulose fractionation are still very limited.[35] Recently, it was reported that alkaline ILs with choline as the cation and amino acids as the anions could selectively extract lignin from lignocellulose.[36] Besides, protonated ILs (PILs) based on the organic superbase tetramethylguanidine (TMG) showed good performance in lignocellulosic biomass processing.[37] Inspired by these results in alkaline DESs and ILs, it was therefore deduced that innovative ADESs could be potentially constructed using suitable organic bases (*e.g.*, choline hydroxide (ChOH) and TMG), which promisingly possessed good capacity for lignocellulose fractionation.

Herein, several novel ADESs (Scheme 1) were prepared based on four HBAs (*i.e.*, imidazole (Im), L-carnitine (LC), tetramethylguanidine (TMG), and choline hydroxide (ChOH)) and three HBDs (*i.e.*, xylitol (Xyl), urea (U), and glycerol (Gly)). It was observed that the constructed ADESs could effectively dissolve various types of lignin at low temperatures (40, 50, and 60 °C), including enzymatically hydrolyzed lignin (EHL), alkali lignin (AL), dealkaline lignin (DAL), and kraft lignin (KL). Moreover, the developed ADESs showed good performances on the pretreatment of corn stover (CS) at 120 °C with a time of 6 h, thus enhancing the subsequent enzymatic hydrolysis to generate sugars. The good capacity for low-temperature dissolution of lignin and efficient fractionation of lignocellulosic biomass confirmed the great potential of the developed ADESs in practical applications.

**Scheme 1. Chemical Structures of the Employed HBAs and HBDs**




## EXPERIMENTAL SECTION

**Construction of the Desired ADESs.** To prepare the desired ADESs, the corresponding HBDs (*i.e.*, Gly, U, or Xyl) and HBAs (*i.e.*, LC, ChOH, TMG, or Im) were first mixed with an HBDs/HBAs molar ratio of 2:1, 1:1, or 1:2. Then, the mixture was heated at 80 °C until a clear liquid was formed. After that, the obtained liquids were dried under vacuum at 60 °C for 24 h. Finally, all the prepared ADESs were stored in a vacuum desiccator for subsequent use.

**Dissolution of Lignin in the Prepared ADESs.** The lignin solubilities in the prepared ADESs were measured based on the previously reported route.[38]

**Fractionation of CS by the Constructed ADESs and Characterization.** In a typical route, the CS and the corresponding ADES with a solid-to-liquid ratio of 1:10 were added to a three-necked flask, and the mixture was heated at 120 °C for 6 h. The detailed steps are given in the Supporting Information.

The chemical compositions of the CS and the generated solid residues were determined according to National Renewable Energy Laboratory (NREL) standard analytical procedures.[39] The Kamlet–Taft parameters of ADESs and the linear correlation of the Kamlet–Taft parameters and lignin solubility were measured and analyzed based on previous literature.[25,38] All characterization methods are given in the Supporting Information.

## RESULTS AND DISCUSSION

**Preparation and Characterization of the ADESs.** As described in Supporting Information, the desired ADESs were prepared by using suitable HBAs and HBDs with different HBAs/HBDs molar ratios (*i.e.*, 2:1. 1:1, and 1:2), and the obtained DESs were named as HBA-HBD-*x*, where *x* represents the molar ratio of HBAs and HBDs (Scheme 1).

To confirm the formation of DESs, we performed $^1$H NMR (nuclear magnetic resonance), FT-IR (Fourier transform infrared) spectroscopic, and DSC (differential scanning calorimeter) experiments. Taking Im-Gly-1:1 as an example, $^1$H NMR spectra clearly showed that the signals for −NH in imidazole and −OH in glycerol moved from 12.14 and 5.04 ppm to 6.34 ppm after the formation of DESs (Figure 1A), implying the generation of strong hydrogen bonds between imidazole and glycerol. Meanwhile, the hydrogen-bond interaction was further confirmed by FT-IR spectroscopy (Figure 1B). In the infrared spectrum of glycerol, the O−H stretching vibration appeared at about 3290 cm$^{-1}$, which moved to 3208 cm$^{-1}$ and became broader after the formation of Im-Gly-1:1, indicating the formation of abundant hydrogen bonds between imidazole and glycerol. In the infrared spectrum of imidazole, the −C═N stretching vibration of imidazole occurred at 1668 cm$^{-1}$, which disappeared after the formation of DESs with glycerol, probably resulting from the intermolecular interaction between imidazole and glycerol. Furthermore, DSC examinations showed that the melting points ($T_m$'s) of the determined ADESs were much lower than those of the pure components (Figure 1C), which was attributed to intermolecular hydrogen-bonding interactions. For example, the $T_m$ values of glycerol and imidazole were 20 and 88 °C, respectively, while the glass transition temperature ($T_g$, −68.9 °C) of Im-Gly-1:1 was far lower than the $T_m$ of the single component. Finally, the thermal stabilities of the prepared ADESs were analyzed. As shown in Figure 1D, the 5% decomposition temperatures ($T_{d5\%}$'s) of the three ADESs (*i.e.*, Im-Gly-1:1, LC-Gly-1:2, and LC-Xyl-1:2) were all above 146 °C (higher than the temperatures for lignin dissolution and CS pretreatment), confirming the relative stabilities of the prepared


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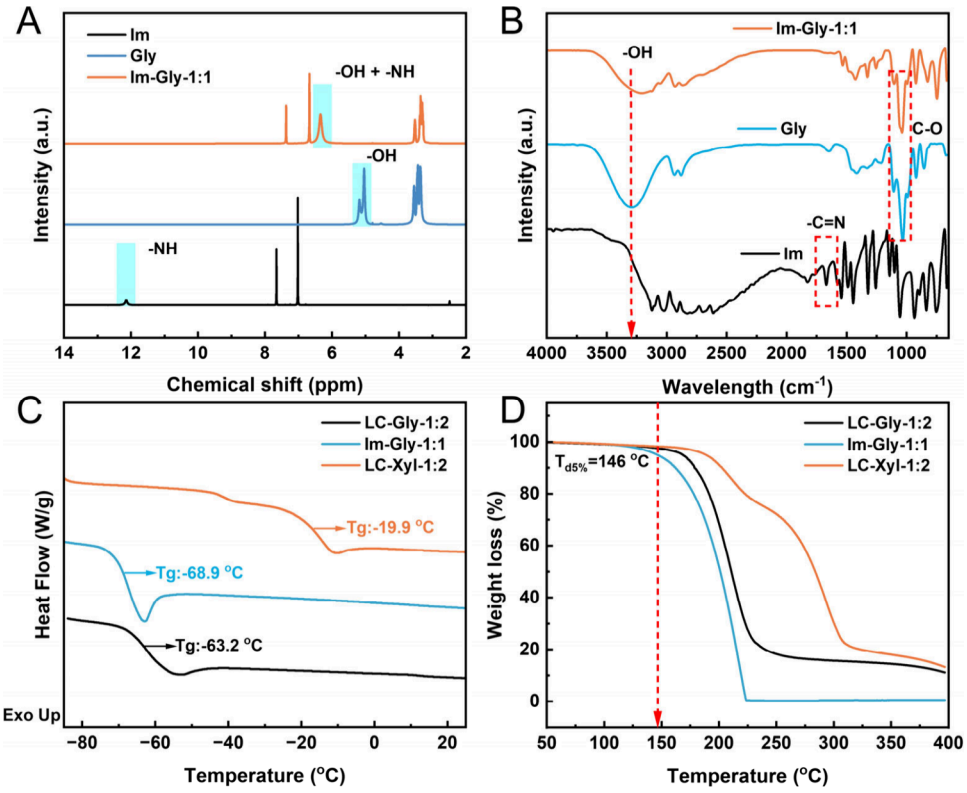



**Figure 1.** (A) $^{1}$H NMR, (B) FT-IR spectra, (C) DSC curves, and (D) TG curves.

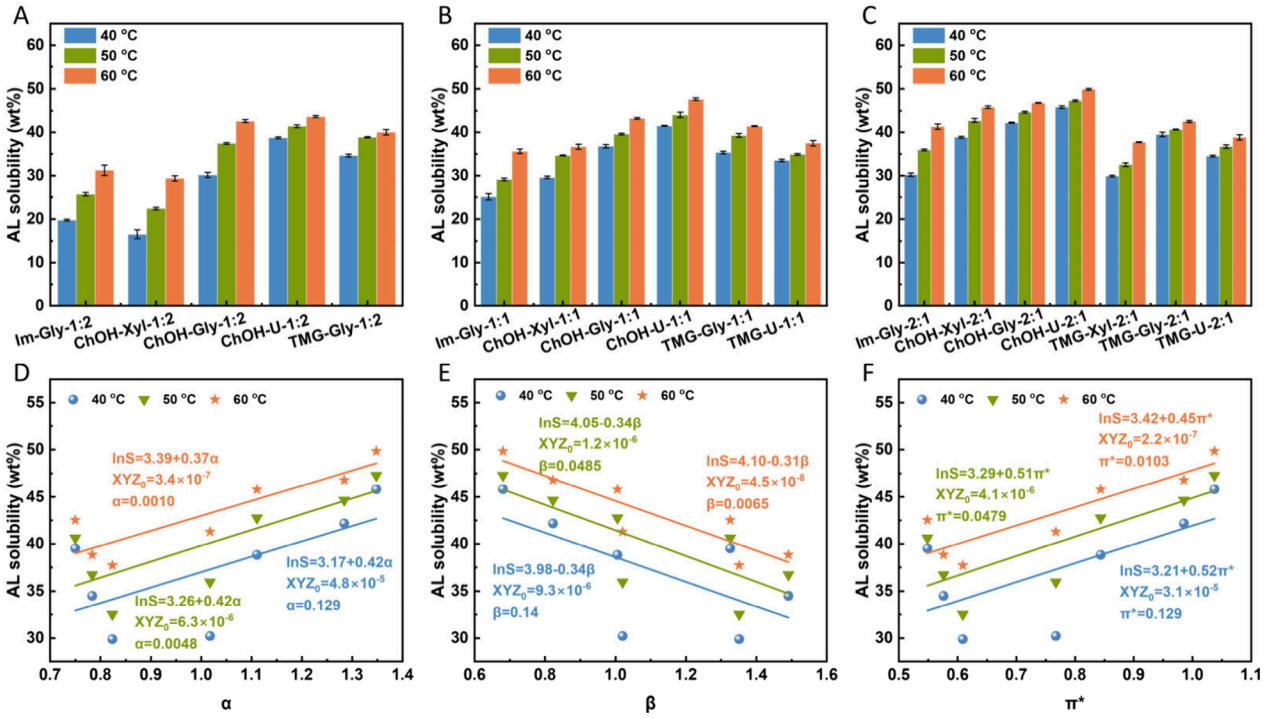



**Figure 2.** Solubility of AL in the prepared ADESs (A–C) and correlations between the lignin solubility and the Kamlet–Taft parameters of the prepared DESs (D–F).

ADESs in the processes of lignin dissolution and CS pretreatment.

**Dissolution of Lignin in the Prepared ADESs.** Among the alkaline DESs used, all of them were liquid at 25 °C (Figure S1). The lignin solubility test temperature was 40–60 °C, proving that all DESs were liquid at the test temperature, and subsequently, alkali lignin (AL) was used to evaluate the capacity of the prepared ADESs to dissolve lignin (Figure 2). First, when ChOH-based ADESs with a molar ratio of 1:2 (*i.e.*, ChOH-Xyl-1:2, ChOH-Gly-1:2, and ChOH-U-1:2) were



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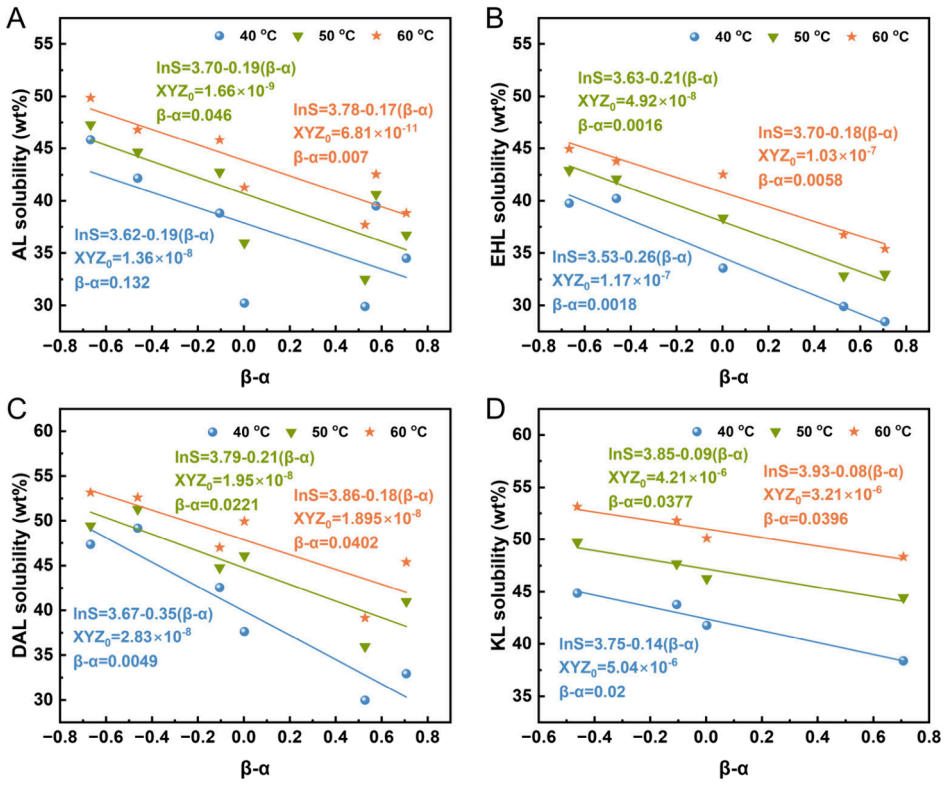



**Figure 3.** Correlations between the AL (A), EHL (B), DAL (C), and KL (D) solubilities and the $\beta - \alpha$ values of the prepared DESs.

examined, ChOH-U-1:2 showed a higher ability to dissolve lignin compared with the other two ADESs. More importantly, the lignin solubility in the used ChOH-based ADESs increased with the increase of dissolution temperature from 40 to 60 °C, and a maximum solubility (49.85 wt %) could be obtained in ChOH-U-1:2 at 60 °C. Besides, when the HBAs/HBDs molar ratio increased from 1:2 to 2:1, the lignin solubility gradually increased, suggesting that more HBAs were beneficial to lignin dissolution. For example, the lignin solubility in ChOH-U-*x* increased from 43.67 to 49.85 wt % at 60 °C when the ChOH/U molar ratio increased from 1:2 to 2:1. Second, TMG-based DESs (*i.e.*, TMG-Xyl, TMG-Gly, and TMG-U) were examined (Figure 2). Notably, TMG-Xyl and TMG-U were unable to form DESs at a molar ratio of 1:2, while TMG-Xly-1:1 was unable to dissolve lignin, because it was solid at the tested temperature. Similarly, the lignin solubility in TMG-Gly-*x* increased with increasing temperature and the TMG/Gly molar ratio. Third, a similar phenomenon was observed for the dissolution of AL in the prepared Im-based ADESs (Figure 2). Finally, after a careful analysis of the results for AL dissolution in these prepared ADESs, it was observed that ChOH-U-*x* showed the highest solubility of AL among all of the developed ADESs when the HBAs/HBDs molar ratio and the dissolution temperature were the same.

Inspired by the good performance of the prepared ADESs on AL dissolution, we attempted to use them to dissolve three other types of lignin (*i.e.*, EHL, DAL, and KL). Similar to the results for AL dissolution, the solubilities of these three types of lignin increased with the increase of the dissolution temperature (from 40 to 60 °C) and the HBAs/HBDs molar ratio (Figure S2). Notably, the best ADES varied with the type of lignin. ChOH-U-2:1 was the best DES to dissolve AL, EHL, and DAL, and their solubility at 60 °C could reach 49.85, 49.46, and 53.17 wt %, respectively. In comparison, ChOH-Gly-2:1 provided the highest solubility of KL (53.14 wt %). These results were probably caused by the different chemical structures of different lignins.

**Correlation of Kamlet−Taft Parameters and Basicity with Lignin Solubility.** As is well-known, the solvation properties of Kamlet−Taft parameters have been quantified to better understand the lignin dissolution behavior[40] in ILs,[41] DESs,[25] γ-valerolactone (GVL)/$H_2O$,[42] and dimethyl isosorbide (DMI)/$H_2O$.[38] $\alpha$, $\beta$, and $\pi^*$ are the most commonly used Kamlet−Taft parameters, which represent hydrogen-bond-donating ability (acidity), hydrogen-bond-accepting ability (basicity), and polarity, respectively. In this study, the $\alpha$, $\beta$, and $\pi^*$ values of the ADESs were determined and analyzed by the linear solvation energy relationship (LSER) statistical method fitted by multiple linear regression (Figure S3).[43] Initially, AL and the ADESs with an HBAs/HBDs molar ratio of 2:1 were selected as the model lignin and DESs (Figure 2D,E). The results showed that the correlation between the AL solubility at 40 °C and the Kamlet−Taft parameters was quite poor. The correlation between the AL solubility and the $\alpha$ value was $\ln S = 3.17 + 0.42\alpha$, $\alpha = 0.129$ ($p > 0.05$). In contrast, good correlation was observed between the AL solubility at 50 and 60 °C and $\alpha$, $\beta$, and $\pi^*$. AL solubility is positively correlated with the $\alpha$ and $\pi^*$ values and negatively correlated with the $\beta$ value. For example, the correlations between the AL solubility and $\beta$ or $\alpha$ values were as follows: $\ln S = 4.10 - 0.31\beta$, $\beta = 0.0065$ ($p < 0.05$), and $\ln S = 3.42 + 0.45\alpha$, $\alpha = 0.0103$ ($p < 0.05$). For other types of lignin (*i.e.*, EHL, DAL, and KL), the relationships between their solubilities and Kamlet−Taft parameters were also analyzed (Figure S3). However, the correlations between the solubilities for these types of lignins and the Kamlet−Taft parameters were not so good as those for AL. Interestingly, positive correlations between the lignin solubility (EHL and DAL) and the values of $\alpha$ and $\pi^*$ would be observed when some


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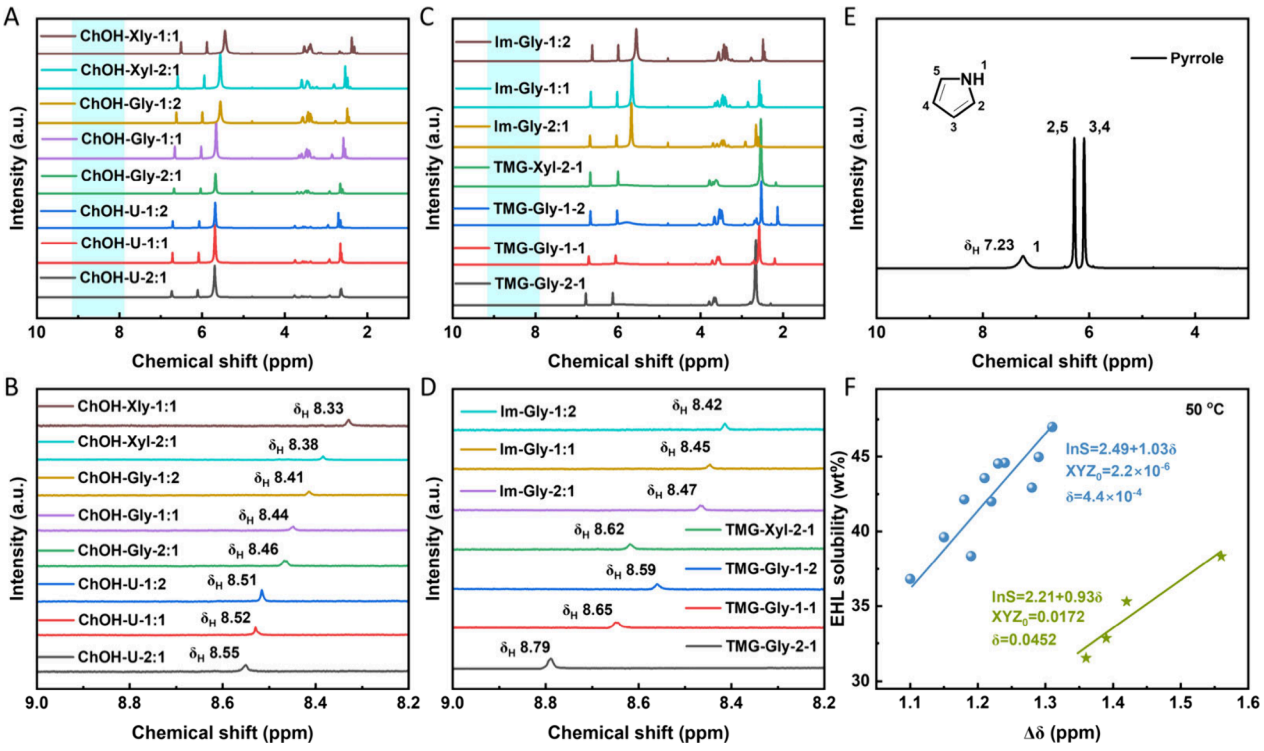



**Figure 4.** [1]H NMR spectra (A and B for pyrrole mixed with ChOH-based DESs, C and D for pyrrole mixed with Im and TMG-based DESs, E for pyrrole) and correlation between EHL solubility at 50 °C and $\Delta\delta$ values of pyrrole in ADESs (F).

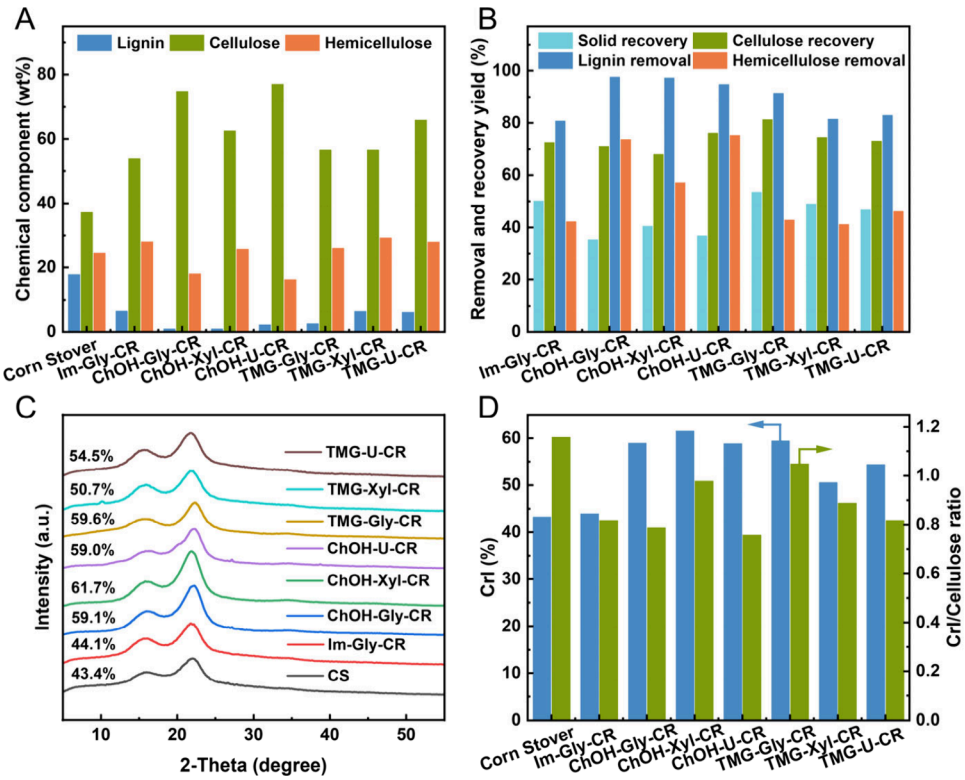



**Figure 5.** Chemical components of various samples (A), recovery of solid and cellulose and the removal of lignin and hemicellulose at 120 °C for 6 h (B), XRD patterns (C), and the crystallinity index (CrI) and CrI/cellulose ratio of corn stover (CS) (D). In this figure, X and CR represent the DES and cellulose-rich solid residues, respectively.

DESs with poor correlation were excluded (Figure 2). Considering that the prepared DESs are basic, the relationship between the lignin solubility and the net basicity ($\beta - \alpha$) of hydrogen bonds of DESs was then explored. As can be seen from Figure 3, lignin solubility showed a negative correlation with the $\beta - \alpha$ value of ADESs. For example, the correlation between four



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lignin samples solubility at 60 °C and $\beta - \alpha$ value were as follows: $\ln S = 3.78 - 0.17\ (\beta - \alpha)$, $\beta - \alpha = 0.007\ (p < 0.05)$; $\ln S = 3.70 - 0.18\ (\beta - \alpha)$, $\beta - \alpha = 0.0058\ (p < 0.05)$; $\ln S = 3.86 - 0.18\ (\beta - \alpha)$, $\beta - \alpha = 0.0402\ (p < 0.05)$; and $\ln S = 3.93 - 0.08\ (\beta - \alpha)$, $\beta - \alpha = 0.0396\ (p < 0.05)$. In other words, the lignin solubility was correlated to the net acidity $(\alpha - \beta)$ of hydrogen bonds of DESs. From the discussions above, it could be inferred that DESs with relatively high $\alpha$ and $\pi^*$ values generally provided higher lignin solubilities. Therefore, $\alpha$ and $\pi^*$ could be probably used as promising indicators for lignin dissolution and extraction by ADESs, which was consistent with previously reported results.[25]

Considering that all of the prepared DESs were basic, it was crucial to reveal the relationship between the basicities of ADESs and the dissolution or extraction of lignin. Although ADESs had been used for lignin extraction and enzymatic saccharification,[33,35] no work has quantified the relationship between lignin solubility and the basicities of the DESs. Here, pyrrole was used as a probe molecule to measure the basicities of the DESs.[44] The formation of hydrogen bonds between the N−H group in pyrrole and the ADESs would cause a lower-field shift of the $^1$H resonance for the N−H group. Especially, the $\delta_H$ value became higher when the ADES possessed stronger basicities. The results for the $\delta_H$ values of the N−H group in the ADESs are provided in Figure. 4A−D, and the basicities of several ADESs could not be given due to high viscosity (TMG-Xyl-1:1 and TMG-U-$x$). It was observed that the N−H peak of pyrrole appeared at 7.23 ppm (Figure 4E). After the pyrrole interacted with the ADESs, the peak of N−H shifted toward 8.33−8.79 ppm (Figure 4A−D). More importantly, the $\delta_H$ value gradually increased with the increase in the HBAs/HBDs molar ratio, indicating that the basicities of the ADESs increased with the increase in the HBAs/HBDs molar ratio. For example, the $\delta_H$ value with ChOH-U-1:2 was 8.51 ppm, and it became 8.55 ppm with the DES of ChOH-U-2:1. Subsequently, the relationship between the EHL solubility and the DES's basicity was analyzed. For Im-, ChOH-, and TMG-based DESs, the EHL solubility was positively correlated with the basicities of the prepared ADESs (Figure 4F), and the correlation between the EHL solubility and the $\Delta\delta$ value was as follows: $\ln S = 2.49 + 1.03\delta$, $\delta = 0.00044\ (p < 0.05)$. For other types of lignin (*i.e.*, DAL, AL, and KL), a similar phenomenon could also be observed for the relationship between lignin solubility and the DES's basicity (Figure S3J−L). From the discussions above, it could be concluded that the DESs with stronger basicities provided higher lignin solubility. In addition, it had been reported that the stronger the basicity of ILs, the better the biomass pretreatment effect.[30,38] It could be deduced that the basicities of DESs could affect the dissolution of lignin and the efficiency of biomass pretreatment. Thus, the basicities of DESs could also be used as one indicator for an ADES's capacity to dissolve lignin and pretreat biomass.

**Pretreatment of Corn Stover by the Fabricated ADESs.** Encouraged by the excellent performance in dissolving various types of lignin, attempts were conducted to use prepared ADESs to remove lignin from lignocellulose. Here, the performances of the ADESs with an HBAs/HBDs molar ratio of 2:1 in the removal of lignin in CS was explored with a solid/liquid ratio of 1:10 at 120 °C for 6 h. Obviously, the composition of CS changed significantly after it was pretreated by the prepared ADESs (Figure 5A). Moreover, the yield (between 35.5 and 53.7%) of solid residues after pretreatment varied with the types of ADESs (Figure 5B). The low yields of solid residues originated from the removal of lignin and hemicellulose in the ADESs during the pretreatment process. The removal efficiency of hemicellulose varied between 41.42 and 75.43%, while that of lignin varied between 80.9 and 97.8% (Figure 5B). In particular, ChOH-Gly-2:1 and ChOH-U-2:1 showed high lignin (97.8 and 95.0%, respectively) and hemicellulose (73.8 and 75.4%, respectively) removal, indicating the effectiveness of the given conditions for the removal of lignin and hemicellulose. It was worth noting that the recovery of cellulose varied between 68.23 and 83.24%, implying that cellulose was partially dissolved during the pretreatment process.[45,46] In addition, the removal effect of ChOH-based DESs on lignin and hemicellulose was better than that of TMG-based DESs (Figure 5B), while TMG-based DESs had a higher retention of cellulose. Briefly, these results above suggested the ability of the prepared ADESs to selectively remove lignin and hemicellulose from lignocellulosic biomass.

Generally, the crystallinity of lignocellulose was changed during the pretreatment process, which was an important parameter for evaluating the efficiency of enzymatic hydrolysis.[29,47,48] As shown in the XRD (X-ray diffraction) patterns (Figure 5C), the CS feedstock and the solid residues after ADESs pretreatment showed two major diffraction peaks (16 and 22°), corresponding to the structure of natural cellulose I. These results indicated that the crystalline structure of cellulose was not altered obviously during the ADESs pretreatment.[28,29] Moreover, the crystallinity index (CrI) of the untreated CS was 43.4% (Figure 5D), while the CrI of the solid residues obtained by ADESs pretreatment increased to 44.1−61.7%. The increase in CrI was attributed to the removal of lignin and amorphous hemicellulose during the ADESs pretreatment. Notably, the CrI values of the solid residues achieved from ChOH-based ADESs were greater than those from TMG- and Im-based DESs, further suggesting that the ChOH-based DESs provided a higher removal of lignin and hemicellulose. Additionally, the CrI/cellulose ratio for the solid residues after ADESs pretreatment was lower than that for the untreated CS, implying that the cellulose content in the solid residues increased much faster than CrI, which was favorable for the subsequent enzymatic hydrolysis.[29,47]

**Enzymatic Saccharification.** The enzymatic digestion performance of cellulose is used to judge the pretreatment efficiency.[28,29,33,49] The untreated CS and the residues obtained from the ADESs pretreatment were used for enzymatic hydrolysis at 60 FPU/g with 2 wt % substrate concentration. When the untreated CS was directly used for enzymatic hydrolysis, the maximum enzymatic hydrolysis saccharification rate for the generation of glucose and xylose was only 14.20 and 18.5%, respectively, with a reaction time of 72 h (Figure 6). During the ADESs pretreatment process, most of the lignin was removed and the complex three-dimensional network structure of lignocellulose was destroyed. Thus, the surface area of the residue significantly increased, which could increase the contact area between the cellulose and the cellulase enzyme, thus promoting the enzymatic hydrolysis efficiency.[33,50] As a result, the sugar yields from the residues obtained by ADESs pretreatment were 5.0−6.0 times higher than that from the untreated CS with an enzymatic saccharification time of 72 h. From these ADES-regenerated residues, the yields of glucose and xylose could reach 71.5−85.4 and 78.5−95.6%, respectively, which were comparable to the enzymatic hydrolysis efficiencies of CS treated by different methods in the literature (Table 1). The results above confirmed that the constructed ADESs were good solvents for biomass pretreatment, which could promote


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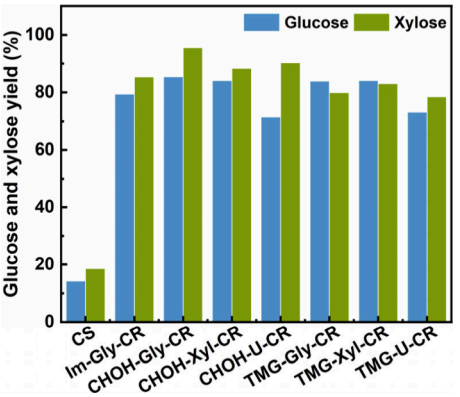



**Figure 6.** Glucose and xylose yields from enzymatic hydrolysis of untreated CS and the generated residues from the pretreatment process in different ADESs.

glucose production from the residues obtained by the ADESs pretreatment.

**Structural Characterization of Lignin.** To elucidate the structural differences in the regenerated lignins from different DESs systems, the chemical structures were analyzed using 2D HSQC NMR (heteronuclear single quantum coherence nuclear magnetic resonance) spectroscopy.[58] Herein, cellulolytic enzyme lignin (referred to as CEL) extracted from identical CS was employed as a representative model of natural lignin, facilitating a comparative assessment of lignin structural changes after the pretreatment process.

It was observed that all of the lignin samples showed similar 2D HSQC NMR spectra except for some minor differences (Figure 7). For the examined lignin samples, the $\beta$-$O$-4 bonding bonds dominated all the 2D HSQC spectra, and a clear correlation signal for the $\beta$ position was observed. Besides, the presence of the $\gamma$-acylated lignin (A′$\gamma$) structure in CEL was attributed to the acylated $\beta$-$O$-4 linkage substructure, which disappeared after the ADESs pretreatment process. Furthermore, the relative abundance ($X$/100Ar) of the different bonds in all lignin samples was quantified by integrating the correlation signals. The content of $\beta$-$O$-4 aryl ether bonds in CEL was 50.5/100Ar, while the relative abundance of $\beta$-$O$-4 aryl ether bonds in ADESs-extracted lignin samples decreased to 8.04−41.7/100Ar, indicating that some $\beta$-$O$-4 aryl ether bonds were cracked during the pretreatment process. More notably, the content of the $\beta$-$O$-4 aryl ether bonds in the lignin samples obtained using ChOH-based DESs was much lower than that using TMG- and Im-based DESs, suggesting that more $\beta$-$O$-4 bonds were cleaved in ChOH-based DESs. Among all the DESs-extracted lignin samples, TMG-Xyl-L had the highest content of $\beta$-$O$-4 aryl ether bonds with a retention rate of 82.3%. The possible reason was that, during the pretreatment process, the hydroxyl group at the C$\alpha$ position of the $\beta$-$O$-4 structure was hydrolyzed to form carbocation intermediates. In the previous acidic DESs pretreatment, the benzene ring of lignin acted as a nucleophile to attack the carbocation intermediates to form condensed lignin.[59] Here, the ADES acted as a nucleophile to attack the carbocation intermediates; the ADES was grafted to the C$\alpha$ position to protect the cracking of lignin, thereby protecting its structure. In the aromatic region, some signals of $p$-coumarate disappeared and the S/G ratio of some samples increased after the ADESs pretreatment, probably due to the preferential degradation of low-molecular-weight G-type lignin over S-type lignin at mild temperature (120 °C).[60] The structural characterizations above indicated that ADESs with high delignification rates can be used to obtain either lignin fractions with a relatively complete structure or fragmented lignin fractions by modulating the type of HBAs.

The average molecular weights ($M_w$ and $M_n$) and polydispersity index (PDI) values of CEL and the regenerated lignin were further analyzed (Table 2). Obviously, the weight-average molecular weight ($M_w$) decreased from 3141 (for CEL) to 2028−3023 g mol$^{-1}$ (for the regenerated lignin from the ADESs systems). More importantly, the molecular weight was positively correlated with the content of the ether bonds as determined by 2D HSQC NMR. In addition, some of the regenerated lignin fractions (*i.e.*, ChOH-Gly-L, ChOH-Xyl-L, and ChOH-U-L) showed homogeneous molecules, which would be favorable for their subsequent value-added applications.

**Mechanism of Dissolving Lignin by ADESs.** The dissolution and delignification mechanism of lignin in ADESs was similar to that of the NaOH pulping method.[61] The HBAs and HBDs in DESs diffused into the structure of lignocellulose through hydrogen bonds, and DESs acted as nucleophilic reagents to disrupt the LCC interactions and promote the solubilization of lignin.[61] When the lignin structure was the phenolic units, the DES acted as a nucleophile and the $\alpha$-ether bond broke to a methylene quinone structure, followed by partial formation of an epoxide intermediate and continuing with the breakage of the $\beta$-ethyl ether bond (Figure 8A).[20,59]

**Table 1. Comparison of the Delignification and Enzymatic Hydrolysis of Corn Stover Using Various Solvent Pretreatments**

| entry | solvents | pretreatment condition | delignification (%) | enzyme hydrolysis condition | glucose yield (%) | ref. |
|---|---|---|---|---|---|---|
| 1 | ethylene glycol−0.6 wt % $H_2SO_4$ (w/w) | 120 °C, 1 h | 80.3 | 3 FPU/g, 72 h | 70.6 | 51 |
| 2 | $p$-TsOH/$H_2O$ | 90% $p$-TsOH, 100 °C, 1.5 h | 83.4 | 15 FPU/g, 96 h | 97.6 | 52 |
| 3 | EtOH/$H_2O$ | 50% EtOH, 220 °C, 1.5 h | 67 | 15 FPU/g, 96 h | 89.5 | 52 |
| 4 | [DBNH][Lev] ILs | 140 °C, 6 h | – | 15 FPU/g, 48 h | 0.49 g/g | 53 |
| 5 | $KMnO_4$−EtOH | 1% (w/w) $KMnO_4$, 40 °C, 1 h; 50% EtOH, 185 °C, 1.5 h | 78.54 | 30 FPU/g, 72 h | 89.45 | 54 |
| 6 | [Bmim]$BF_4$/$H_2O$ | 150 °C, 5 h | – | 25 FPU/g, 48 h | 81.68 | 55 |
| 7 | ChCl/EG/$Na_2CO_3$ | 120 °C, 3 h | 88.1 | 15 FPU/g, 72 h | 94.5 | 56 |
| 8 | ChCl/EG/$NaHCO_3$ (2.0 wt %) | 140 °C, 3 h | 90.03 | 60 FPU/g, 72 h | 97.47 | 57 |
| 9 | corn stover (CS) | – | – | 60 FPU/g, 72 h | 14.2 | this work |
| 10 | ChOH/Gly DESs | 120 °C, 6 h | 97.8 | 60 FPU/g, 72 h | 85.4 | this work |



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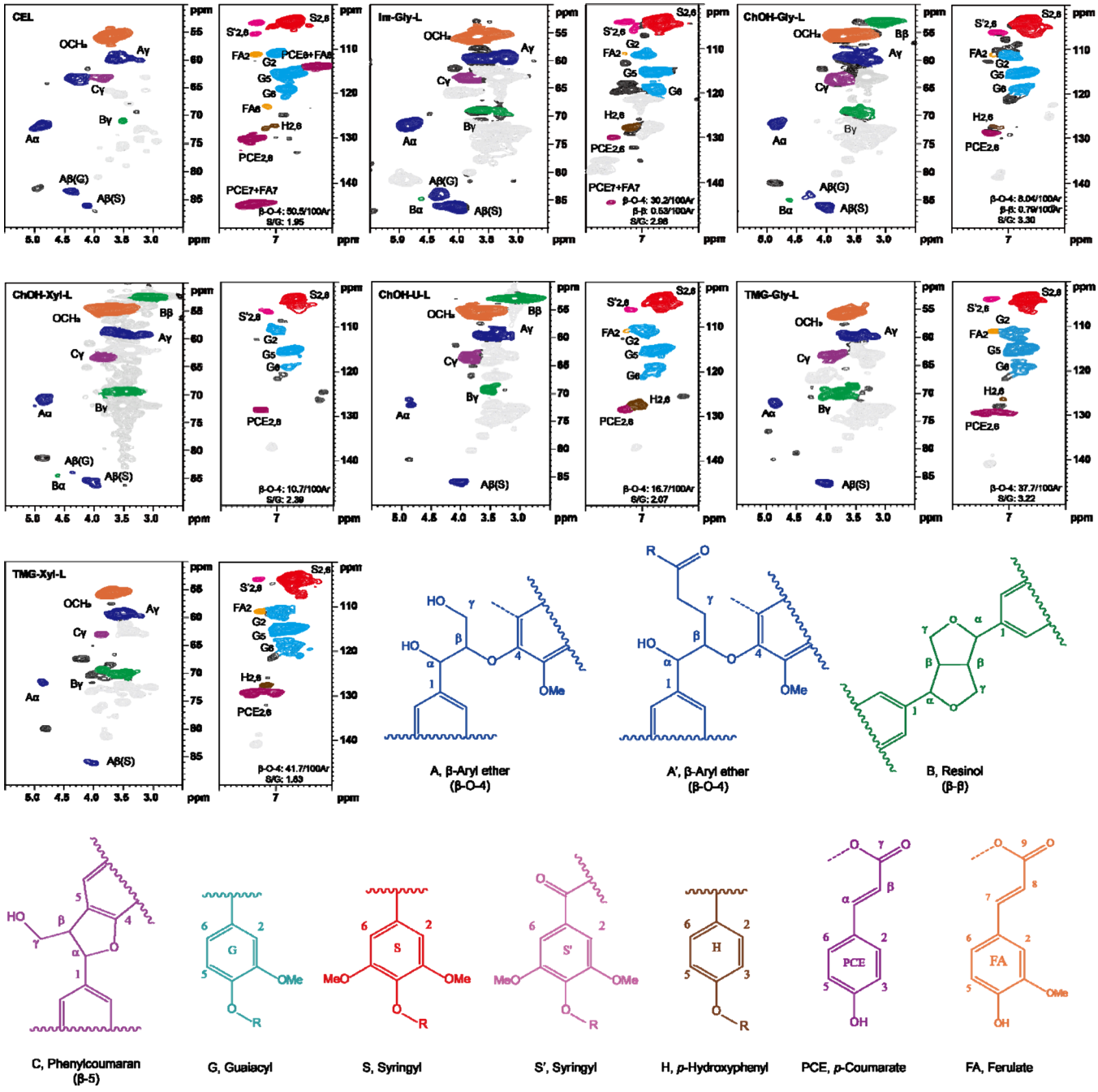



**Figure 7.** Side-chain and aromatic regions in the 2D HSQC NMR spectra of CEL and the regenerated lignin from different ADES systems and the identified substructures in lignin.

**Table 2. Relative Abundance ($X$/100Ar) of $\beta$-$O$-4, $\beta$–$\beta$, and the Molecular Weight of Lignin after the ADESs Pretreatment Process**

| | lignin samples | | | | | | |
|---|---|---|---|---|---|---|---|
| | CEL | Im-Gly-L | ChOH-Gly-L | ChOH-Xyl-L | ChOH-U-L | TMG-Gly-L | TMG-Xyl-L |
| $\beta$-$O$-4 | 50.5 | 30.2 | 8.04 | 10.7 | 16.7 | 37.7 | 41.7 |
| $\beta$–$\beta$ | – | 0.53 | 0.79 | – | – | – | – |
| S/G | 1.95 | 2.98 | 3.30 | 2.39 | 2.07 | 3.22 | 1.83 |
| $M_w$ | 3141 | 3023 | 2079 | 2028 | 2043 | 2868 | 2898 |
| $M_n$ | 1892 | 1411 | 1507 | 1459 | 1570 | 1365 | 1327 |
| PDI | 1.66 | 2.14 | 1.38 | 1.39 | 1.30 | 1.88 | 2.18 |

When the lignin structure was non-phenolic units, the $\alpha$-hydroxyl group dissociated first, and the generated $\alpha$-alkoxy intramolecularly attacked the $\beta$-carbon. Then the $\beta$-$O$-4 bond was broken to generate an epoxy intermediate (Figure 8B).[59,62]



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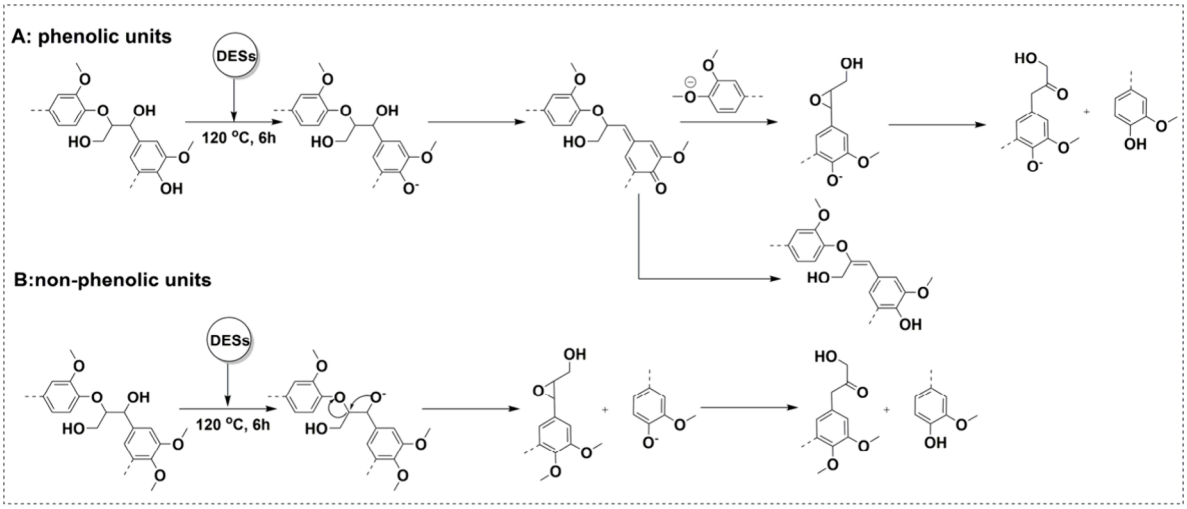



**Figure 8.** Proposed mechanism of dissolving lignin by ADESs: (A) phenolic units; (B) non-phenolic units.

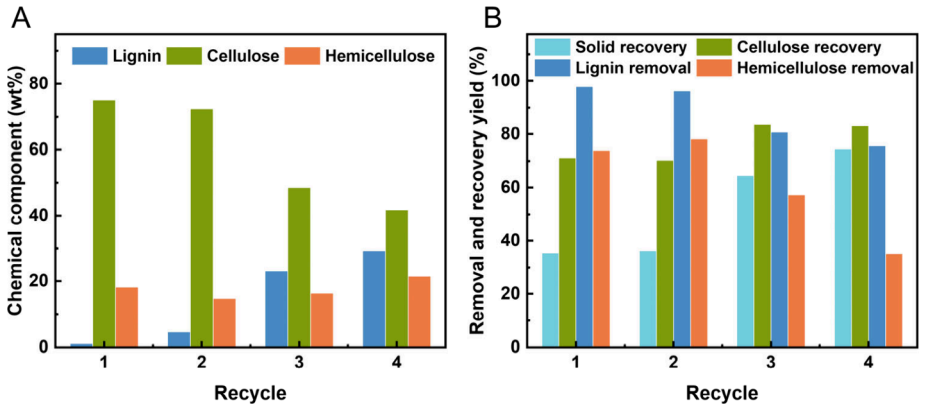



**Figure 9.** Chemical components of samples recycled three times by DESs (A); recovery of solid and cellulose and removal of lignin and hemicellulose at 120 °C for 6 h (B).

**Recovery of DESs.** Solvent recycling is of vital importance in biorefineries. In this work, the recovery of DESs solutions was investigated (Figure S4). It is worth noting that the lignin precipitation process inevitably led to salt production with ChOH due to the addition of HCl to the DES−lignin solution. Therefore, ChOH-Gly DESs were selected for the recycling test. Since ChOH contained water, the masses of DESs were calculated after recycling and the water contents of the recovered DESs were measured by a Karl Fischer hydrometer (Table S3). A certain amount of water was added before the next use to ensure the same water content as the fresh DES. From the results of the component analysis (Figure 9A), after the DESs were reused once, the lignin removal rate (96.2%) was comparable to that of the fresh DESs (97.8%; Figure 9B). After three times of reuse, the removal rate of lignin was still able to reach 75.6%, which was higher than the delignification rate of many DESs when used for the first time,[63−66] proving the excellent recovery performances of DESs.

## FUTURE PERSPECTIVE

Although recovered DESs without treating the proton salts produced could also have a good effect in the subsequent pretreatment, the removal of proton salts was still worth studying in order to recover DESs with the same compositions as fresh DESs. Here, the anion exchange resin method was tried to remove $Cl^-$ (Figure S5) and the dialysis method was used to recover lignin. Both methods can finally recover high-purity DESs. However, these two methods also have their own disadvantages. The cost of the anion exchange resin was not advantageous, and the dialysis method required a large amount of water; in the subsequent process of recovering DESs, the water needed to be removed, which was energy-consuming. Supercritical fluids (for example, carbon dioxide) antisolvent techniques might be useful for the precipitation of lignin from DESs because supercritical fluids have shown the ability to precipitate polymers from solvents. However, we currently do not have an appropriate apparatus to test this idea and hope someone could prove it in the future. Therefore, it is necessary to further find new methods and new antisolvents for precipitating lignin in alkaline DESs to control cost issues and avoid the production of byproducts.

## CONCLUSIONS

In conclusion, several novel ADESs were prepared that showed good performances in lignin dissolution and biomass pretreatment. The lignin solubility positively correlated to the Kamlet−Taft parameters and the basicities of the ADESs. Moreover, the ADESs showed excellent delignification capacities (80.9−97.8%), and the pretreatment of CS by these ADESs could enhance the enzymatic efficiency of the residues (71.5−85.4%). Notably, the pretreatment of biomass could generate lignin with relatively intact structures or lignin fragments with a low


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molecular weight and good dispersion. The prepared ADESs are potential solvents for the efficient fractionation and saccharification of biomass.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssusresmgt.4c00070.

Detailed steps such as materials, dissolution of lignin in the prepared ADESs, fractionation of CS by the constructed ADESs, recovery of DESs, enzymatic hydrolysis, analytical methods, characterizations of lignin and solvent, determination of the Kamlet−Taft parameters of the prepared ADESs, Kamlet−Taft LSER method (PDF)

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### Notes

The authors declare no competing financial interest.

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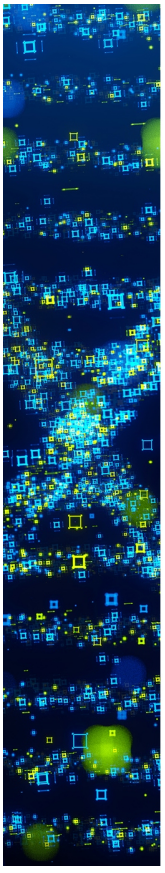
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